Updated: Nov 10, 2025

Update: Supporting Trusted Review Metadata in Harvard Dataverse Repository

Project Timeline

2025					2026						
Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul
x	x	x	11/10								

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Project Goal

Our goal is to design and implement Dataverse platform features and functionality to support machine-readable, community-defined Trusted Review metadata signaling fitness for reuse, provenance, sensitivity, or other domain-specific indicators of trust. These metadata will leverage existing schema.org, DataCite, and Croissant metadata elements to best effect and will be exportable from the Dataverse platform using its API, its metadata export functionality, or another appropriate mechanism.

The project builds on the Dataverse platform's infrastructure and its extensive global network of open science repositories to contribute both metadata standards and production-grade tools that can help communities express what trust means in their specific domain.

Progress Summary

This document describes the Dataverse team's progress from August 2025 through October 2025 towards implementing support for **Trusted Review** metadata in the Harvard Dataverse Repository.

Trusted Review metadata is structured, computer-readable metadata statements co-designed with research communities to signal fitness for reuse, provenance, sensitivity, or other domain-specific indicators of trust. This metadata will be shared using existing formats, such as schema.org, CroissantML, and DDI-CDI, to ensure broad interoperability. The final versions of these metadata will be provided by our use case partners at New York University and the University of North Carolina. Likewise, how NYU and UNC define their scientific communities or communities of interest will inform our definition of **Trusted Review Community** and our expectations for how future applicants define their communities and their expert reviewers' relationships to those communities.

Trusted Reviews (TR), like research datasets, support custom metadata and documentation, appear in Harvard Dataverse Repository searches, have persistent identifiers (e.g., digital object identifiers), can be exported from or harvested from Harvard Dataverse, and can be grouped into Dataverse collections. Trusted Review objects connect a community's trust metadata and assessment scores to the specific dataset or other resource the community has reviewed. The Trusted Review functionality extends Dataverse's underlying support for different resource types, including research datasets, computational workflows, software, and documents.

Initially, we called this functionality 'Trusted Data' and we expected that the metadata would refer to research datasets stored on the Harvard Dataverse platform. However, while gathering requirements from our use case partners, it became clear they needed to assess and assign metadata trust scores to visualizations, summary tables, and other Web-accessible resources, as well as to datasets hosted anywhere. Our terminological shift from 'Trusted Data' to 'Trusted Review' reflects this change in scope and functionality.

Our conversations with our use case partners also reinforced the need for Trusted Reviews (and their metadata) to be API-accessible, not only via the Harvard Dataverse Repository's graphical user interface (GUI), the Dataverse software export functionality, or Web searches or other applications. Our design decisions and proposals reflect this key use case requirement; we will extend existing Dataverse APIs as needed.

To date, we have:

- Collected and analyzed known functional and metadata requirements from our use case partners.
- Developed and presented a prototype and demo of the Trusted Review functionality using sample metadata.
- Identified an appropriate DataCite resourceType, relationTypes, and schema.org metadata category for Trusted Reviews.
- Identified different scenarios to support allowing members of use case communities to assess resources, such as collecting and tabulating assessments and assigning final scores across trust metadata elements (see Community Resource Assessment and Scoring Scenarios below).
- Developed technical designs for implementing the new Trusted Review resource type in Harvard Dataverse (see Trusted Reviews and Metadata Standards).
- Begun user interface design for creating Trusted Review resources.
- Begun developing policies for hosting Trusted Reviews on the Harvard Dataverse Repository (see Hosting Trusted Reviews on Harvard Dataverse).

Next, we plan to:

- Continue gathering functional and metadata requirements from our use case partners, and iterate on our technical design as needed (e.g., input on whether we should support icons or colors in TR metadata), and prototype custom metadata blocks in Harvard Dataverse that reflect our use case partners' trust metadata.
- Gather metadata-sharing requirements (e.g., CroissantML, schema.org) from our Google partners (See Trusted Review Metadata Sharing in the Open Questions sections). Then, design and implement metadata-sharing functionality in accordance with the requirements. We'd like to set up a technical meeting to discuss these details.
- Implement, test, and merge new Trusted Review functionality into the Dataverse codebase.
- Explore technical requirements for scenarios to collect trust assessments (see Community Resource Assessment and Scoring Scenarios) and assign final trust scores on the platform itself.

- Finalize the design and begin implementing the Trusted Review user interface.
- Further develop Harvard Dataverse Trusted Review hosting policies.
- Get input from our project partners about the proposed project Website or blog post mentioned in the original project plan and at the kick-off meeting in August 2025.

Use Case Requirements

During multiple meetings from August 2025 through October 2025, we have gathered functional and metadata requirements from our New York University and University of North Carolina use case partners. Known requirements include:

- Use case representatives may create Trusted Reviews and make trust assignments via the Harvard Dataverse user interface or via Dataverse APIs.
- Reviewed resources may or may not be research datasets. These resources may or may not reside on Harvard Dataverse.
- Use case communities may uncover different standards for trust measures, necessitating different custom metadata blocks.
- Google Dataset Search and other apps and tools must be able to locate and collect information about resource trust assignments (e.g., via the Dataverse metadata export functionality or Dataverse APIs).
- Scholars, researchers, journalists, and others can search for and cite reviews of specific resources and Trusted Review (TR) versions.

Our analysis of the requirements pointed the way to designing a new Dataverse resource type to which we can attach their trust metadata and link to any Web-accessible resources they want to assess (and assign trust metadata), which we currently call Trusted Reviews.

Trusted Reviews

Trusted Reviews are analogous to academic book reviews. They document a community's critical assessment of a visualization, summary table, dataset, or another Web-accessible resource. In the Harvard Dataverse Repository, Trusted Reviews are containers for use case-specific metadata, documentation, files, code, additional metadata, and other optional elements. The Trusted Review functionality leverages existing Dataverse functionality supporting resource types beyond datasets, including software and computational workflows.

Trusted Review Characteristics

Like Dataverse research datasets, Trusted Reviews are citable, have unique persistent identifiers (DataCite-registered DOIs), are versionable, searchable, can be shared via Dataverse

APIs, and are exportable and harvestable (see Figure 1). They enable our use case partners to link Web-accessible resources that their communities have assessed with metadata recording the resources' trust scores. There is **one Trusted Review object for each community-reviewed resource**. Our partners or their users may engage with or share these TRs via Dataverse APIs or the Harvard Dataverse user interface.

Trusted Review Metadata (in Figure 1) refers to the use case-specific fitness-for-use metadata that our partners will provide. A Trusted Review's metadata records the final scores resulting from the community assessments.

Figure 1: Trusted Review Overview

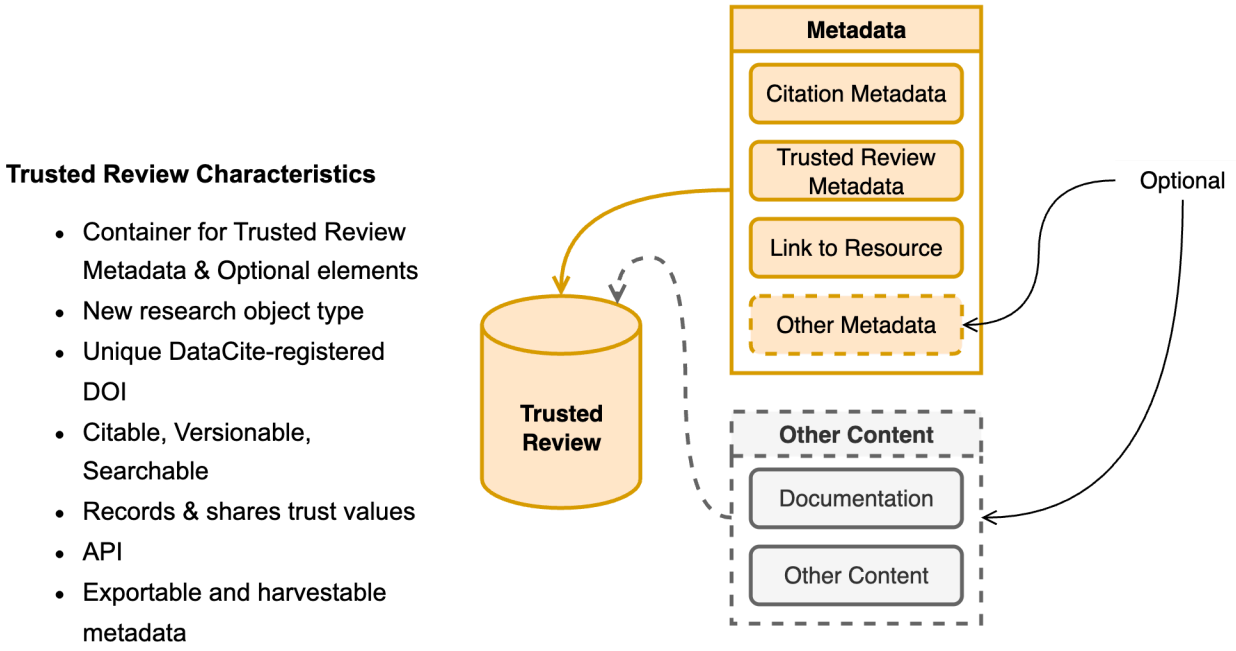
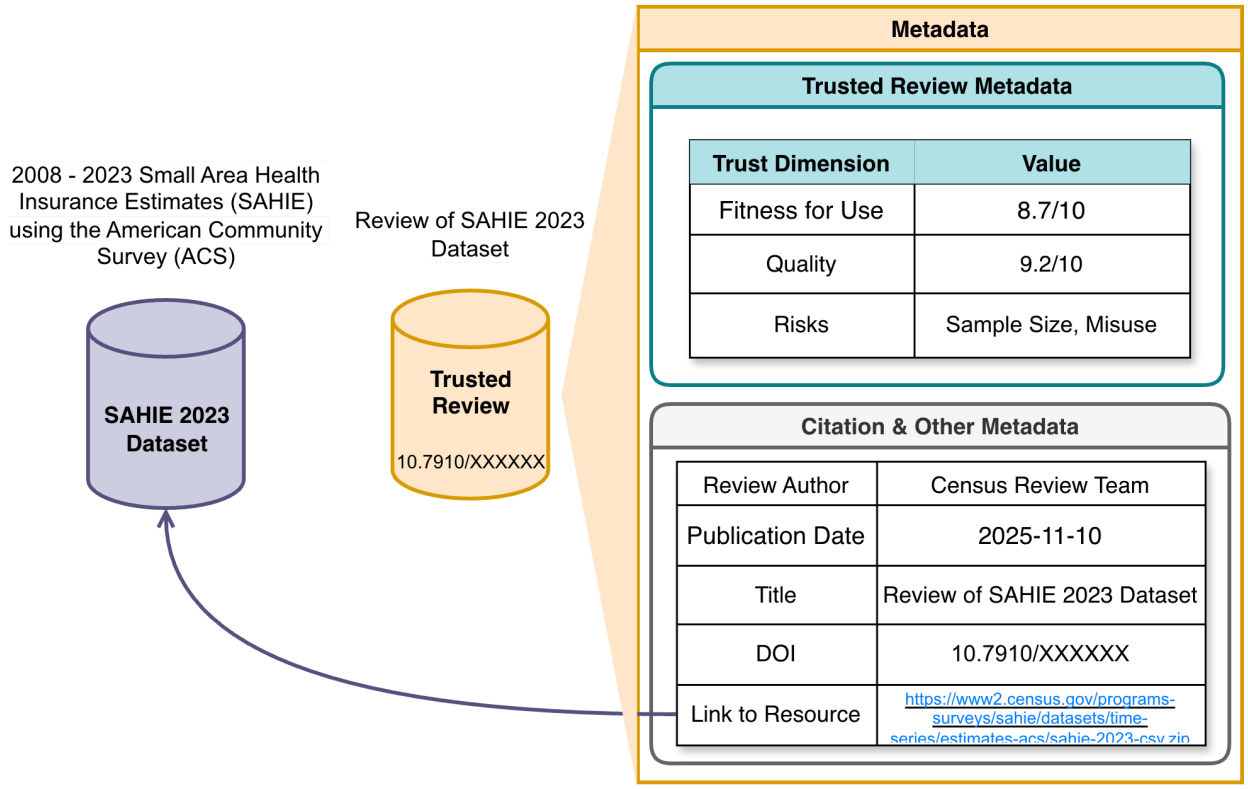


Figure 2 shows the anatomy of a Trusted Review of a U.S. Census dataset. This architecture **enables multiple communities to assess and assign different fitness-for-use metadata scores to the same resources**.

Figure 2: Trusted Review U.S. Census Dataset Example



Trusted Reviews and Metadata Standards

In the Harvard Dataverse Repository, a Trusted Review schema.org CriticReview is:

> "[A] more specialized form of Review written or published by a source that is recognized for its reviewing activities. These can include online columns, travel and food guides, TV and radio shows, blogs and other independent Web sites. CriticReviews are typically more in-depth and professionally written" (schema.org, n.d.).

DataCite supports many *resourceTypes*, including 'Dataset', 'Report', and 'PeerReview'. However, currently CriticReview is unsupported. Therefore, Trusted Review objects will use an 'Other' DataCite resourceType. We are discussing other possibilities with DataCite.

Regardless, for 'Other' resourceTypes, DataCite supports key *relationTypes*, a vocabulary for describing how resources are related to one another. We can use these relationTypes to connect Trusted Reviews with the visualization, dataset, or other resource they refer to, including:

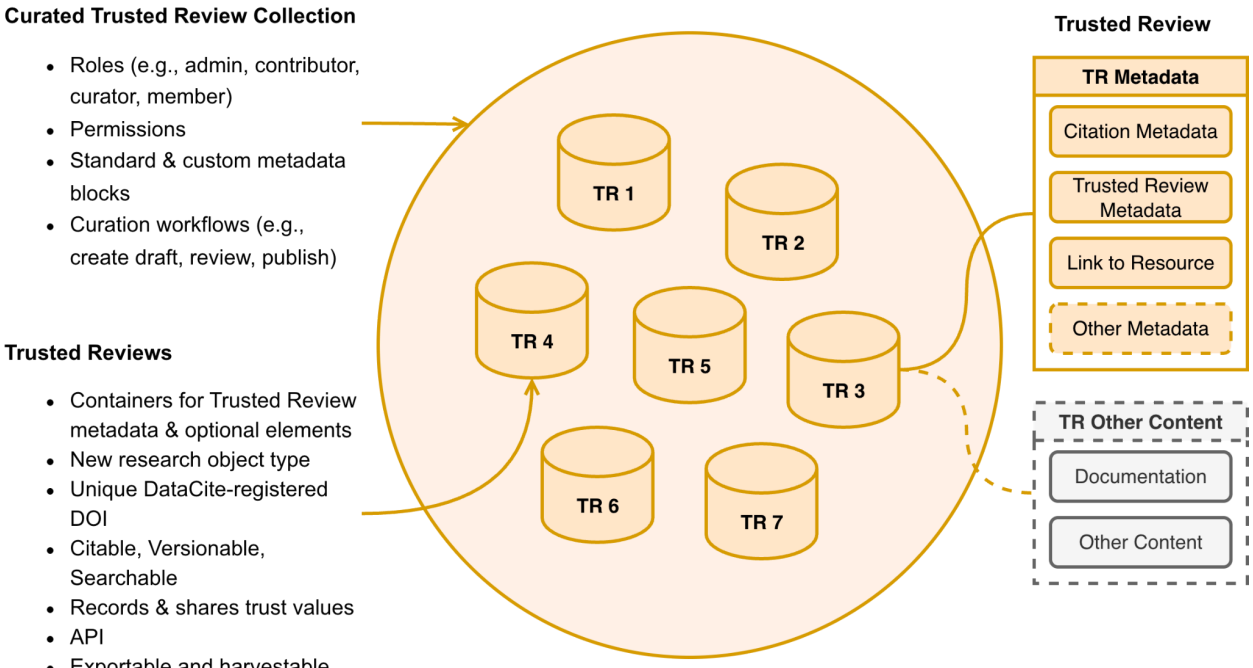
- Reviews: "indicates that A is a review of B"
- IsReviewedBy: "indicates that A is reviewed by B"
- IsMetadataFor: "indicates additional metadata A for a resource B", and

- HasMetadata: "indicates additional metadata A for a resource B"

Trusted Review Collections

Each of our use partners will have a curated collection on Harvard Dataverse that stores Trusted Reviews about the resources their communities have reviewed. Dataverse Collections are containers for Trusted Reviews, datasets, software, and subcollections. Figure 3 shows the anatomy of a collection containing 9 Trusted Reviews, each linked to a separate Web-accessible resource, such as a dataset or a visualization.

Figure 3: Collections of Trusted Reviews on Harvard Dataverse

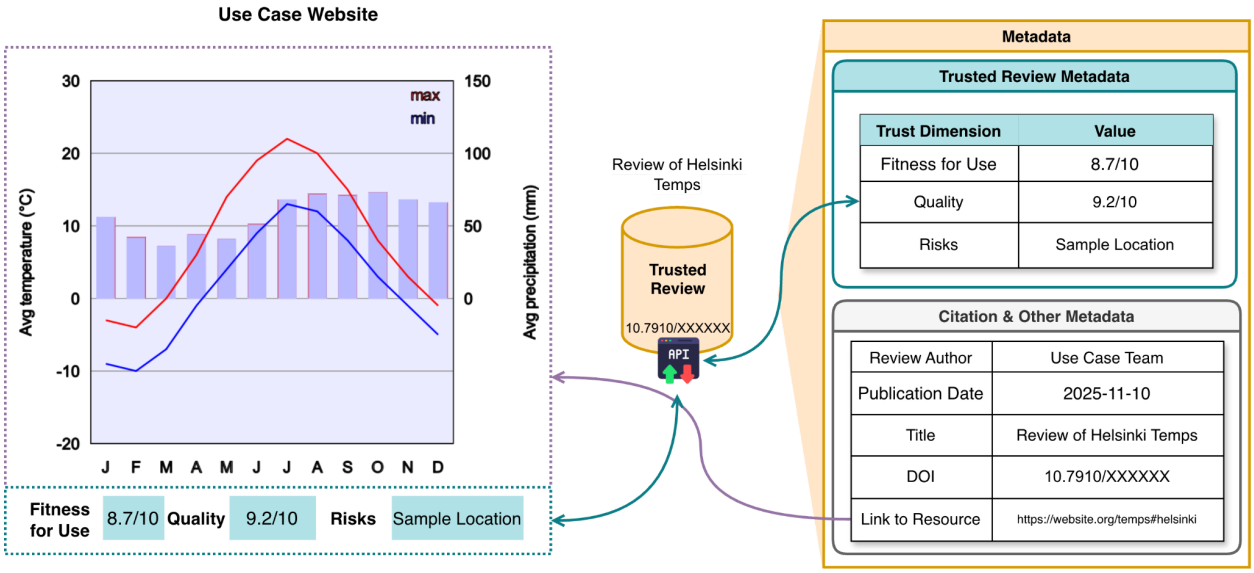


Collection Features and Benefits

Dataverse collections support granular permissions for collection users, draft review creation, managed workflows (e.g., asynchronous submit-for-review workflows), and access via Dataverse APIs to create and manage their contents. For example, the UNC team can manually (via the UI) or programmatically (via the API) create new Trusted Review drafts in their collection, assign metadata scores, add documentation and other metadata, and publish them. Using Dataverse collections will also enable use case teams to customize their TR metadata, share documentation, and links to their organizations, communities, assessment methods, and other resources of interest to their user communities.

Once published, a Trusted Review and its metadata become part of the ever-growing research data and scholarly resources ecosystem. It becomes findable, accessible, interoperable, and reusable (FAIR) (Wilkinson et al., 2016); its metadata and contents (e.g., documentation) can be updated to meet community needs and values, and each version can be cited by others. Trusted Review metadata can also be retrieved via API, enabling an external application to embed trusted metadata scores in its user interface or use the values internally. Figure 4 illustrates how Trusted Review metadata could be retrieved and used via the API, then displayed on a community's Website.

Figure 4: Retrieving and Displaying Trusted Review Metadata Values Via API



Plot credit: OpenClipArt.org

Trusted Review Architecture

Flexibility

Our Trusted Review architecture can accommodate different ways communities may want to create, manage, and share trust metadata scores and link this metadata to resources they've assessed. For example, a community-assessed resource may be hosted by Harvard Dataverse (e.g., a local dataset) or on the Web. Alternatively, the community might have a collection of reviews of datasets that are stored on the Harvard Dataverse or in other research data repositories. Furthermore, community representatives may create Trusted Reviews manually or using a script (e.g., Python, Jupyter Notebook, or a Unix shell) that creates TRs and assigns metadata via the Dataverse API.

User Engagement Modes

From a user engagement perspective, a community's end users may search for and view its Trusted Reviews using the Harvard Dataverse user interface. Or end users may be uninterested in engaging with the Trusted Review as a scholarly object (e.g., citing it in an article) and only ever see final review scores on a community-designed and hosted Website (e.g., see Figure 4).

Custom Trust Metadata

Our approach also enables communities to define their own custom fitness-for-use metadata elements and values. By decoupling fitness-for-use metadata from the review object, we enable multiple communities to review the same resource, assign different fitness-for-use values, and share community-specific metadata with others.

Community Resource Assessment and Scoring Scenarios

The relationship between how communities assess resources and assign scores and Trusted Reviews in collections on Harvard Dataverse is important and worth exploring in greater detail. During our conversations with our use case partners, three scenarios for supporting resource assessment and scoring emerged.

Scenario 1

Initially, we envisioned that a community leader would have community members assess a set of datasets or other resources entirely separate from the Harvard Dataverse platform using a separate process or third-party tool. For example, we expected that a community would convene a panel of experts, have them review a list of resources using a community-defined rubric, either via a Qualtrics survey form or a structured interview, then tabulate the results and record the final scores in the metadata of individual Trusted Reviews on Harvard Dataverse. Scores could be entered manually or uploaded via API. (Currently supported)

Scenario 2

Communities may also prefer to use an external tool, such as a Shiny app, to collect and record their dataset assessments. The tool would collect expert assessments, tabulate them, and write the final scores to metadata in Trusted Review objects in a Harvard Dataverse collection. (Requires application development, but leverages APIs)

Scenario 3

Finally, communities might want to perform the assessment and scoring process on the Dataverse platform itself. Here, use-case-related experts would record their assessments on Trusted Review drafts using brand-new functionality, and a review collection manager would record the final scores in the metadata of the review objects in a Harvard Dataverse collection. (Requires extensive Dataverse codebase modification)

Scenario 1 is currently supported in Harvard Dataverse. We are investigating the technical feasibility of supporting the others. Scenario 2 would require the design and development of a lightweight external tool to perform the assessment and scoring processes and to update Trusted Reviews' metadata. Scenario 3 requires extensive policy and technical analysis before we decide whether and how to implement support for it. Completing Scenarios 2 and 3 is unlikely within the current project timeline.

Hosting Trusted Reviews on Harvard Dataverse

Individual communities may choose not to surface the scholarly characteristics of their Trusted Reviews (e.g., persistent identifiers, versioning, citability). Nevertheless, they are hosted on the Harvard Dataverse. They must therefore adhere to existing repository policies and conform to the standards expected of all FAIR data repositories and other research infrastructures. Our work to establish policies for supporting future Trusted Review use cases on Harvard Dataverse is informed by our current partners' different approaches to defining and engaging with their communities and defining and assessing resources.

Currently, we have identified eight high-level steps in the Trusted Review collection application, approval, implementation, and maintenance process, described below and shown in Figure 5 and in the TR Hosting Workflow tab. The Use Case (UC) Team refers to representatives of future communities who want to host their Trusted Reviews on Harvard Dataverse.

1. **Prepare and Submit Application**. The Use Case (UC) Team prepares and submits or revises a submitted TR hosting application.
2. **Review Application**. The application is reviewed by the Harvard Dataverse Screening Committee. At a minimum, we anticipate that a successful application will contain the following elements:
    a. Clearly defined community.
    b. Description of, and defined selection criteria for expert reviewers.
    c. Clearly defined relationship of expert reviewers to the community.
    d. Clearly defined resource selection criteria.
    e. Assessment tool used by expert reviewers to evaluate resources.
    f. Clearly defined assessment and scoring methodology.
    g. Metadata elements and values to be encoded in a custom Trusted Review metadata block, and a
    h. Sustainability plan.

3. **Create TR Collection and Custom Metadata**. If the application is approved, the Harvard Dataverse curation team creates the TR collection and its custom metadata block on behalf of the UC Team.

4. **Conduct Assessments and Calculate Scores**. Meanwhile, the UC Team assembles its resources, recruits experts in accordance with its application plan, conducts expert assessments, and calculates final trust scores for each metadata element.

5. **Customize Collection and Create TR Drafts**. Simultaneously, the UC Team may also customize their Harvard Dataverse collection (e.g., apply branding, link to a community Website, or add documentation) and create TR drafts. The UC team may create TR drafts one by one using the Harvard Dataverse graphical user interface or via a script that uses the Dataverse APIs. One TR draft is created per resource the UC Team plans to assess.

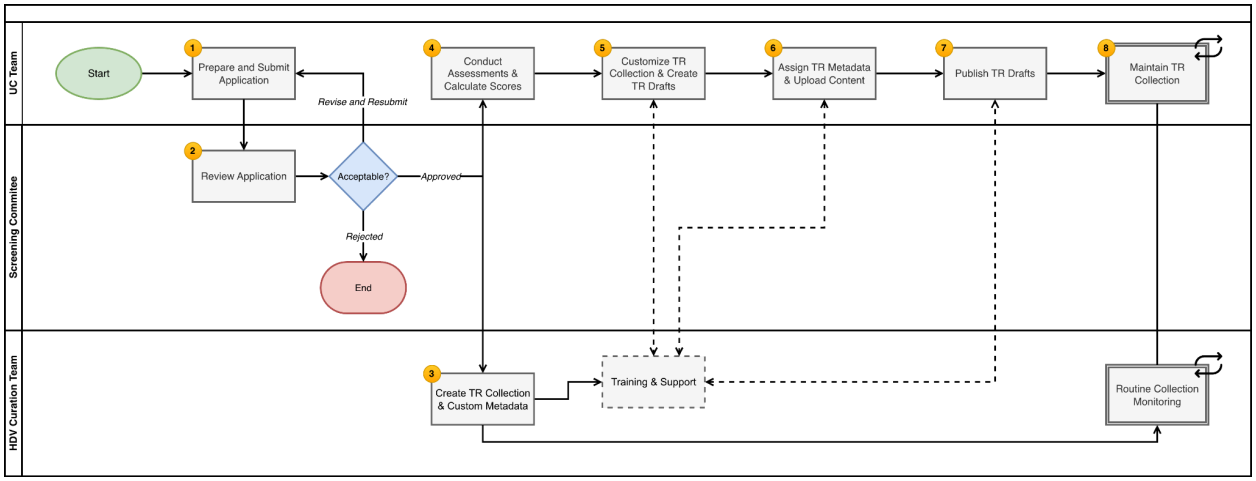
6. **Assign TR Metadata and Upload Content**. Once the assessment phase is complete and the TR drafts are created, the UC Team assigns the final trust scores to each Trusted Review, then uploads any additional content (e.g., READMEs, the assessment instrument) to the collection or TRs.

7. **Publish TRs**. When the TR drafts are complete, the UC Team publishes the Trusted Reviews, making them findable in searches, harvestable, and available via the API to other applications. Upon publication, a unique DataCite DOI is assigned to the Trusted Review, and the TR becomes publicly available. It can be searched for and viewed by anyone via the Harvard Dataverse user interface. Its metadata can also be retrieved from the Dataverse API and shown on a UC Team's website, dashboard, or other application.

8. **Maintain TR Collection**. In the final step, the UC Team continues to maintain its TR collection in accordance with the sustainability plan defined in its application. For example, UC Team members may update TR metadata to reflect new versions of sources, add Trusted Reviews of new resources, or update other documentation.

We plan to iteratively refine this workflow and its processes and to prepare a best-practices and policy guide upon completing the project.

Figure 5: Trusted Review Hosting Workflow



Open Questions

The following questions have arisen during our requirements gathering, design, and prototyping phases and require input from our project partners.

Naming: Trusted Reviews or Something Else?

In August, as a group, we discussed the concept of 'Trusted Data'. However, the NYU use case team identified the need for us to consider the trustworthiness or fitness-for-use of resources beyond datasets, including visualizations and summary tables. Internally, we have been using the term **Trusted Reviews**. But the Dataverse team members have brainstormed other options, such as:

- **Community Reviews**. Emphasizes the community-specificity of the reviews, their metadata characteristics for fitness-for-use, and the scoring.
- **Fitness for Use Reviews**. Emphasizes that trustworthiness is linked to a resource's fitness for use within a specific community or for a specific purpose.

What are everyone's thoughts about these alternatives? Should we be considering others?

Trusted Review Metadata Sharing

We have identified an appropriate schema.org metadata type for Trusted Review objects, along with their corresponding DataCite resourceType. We can also create custom metadata blocks in Harvard Dataverse that correspond to the (known) trusted metadata elements our use case partners need. Next, we need input from our Google partners about their expectations for how we should package and share this metadata.

We have several technical questions about metadata packaging and sharing. For example, CroissantML is intended for datasets; Trusted Review objects are similar to datasets, but contain more metadata.

- Is CroissantML still the preferred packaging and sharing format?
- If so, does the Google team have preferences for how we encode additional trust metadata elements in the Croissant metadata?
- Were you planning to use the Dataverse metadata export function to gather Croissant metadata from Trusted Reviews?
- Or is an API preferred?

We would like to schedule a meeting to discuss these and other technical questions.

Project Website or Blog Posts

Finally, we need input from all of our project partners about the proposed project Website or blog post mentioned in the original project plan. We need input from partners on the site's audience, location, content, and a timetable.

- Should we host the Website or blog post locally, at IQSS, or do you all envision a more enduring location, such as Medium or Substack?
- Or is there another preferred location?
- When should this site go live?

References

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Trusted Review Hosting Workflow

